

Code for CodeHS Heatmap:

Setting up a CodeHS Workspace:

1. Type “CodeHS” into your browser and select the link. Alternatively, click the link [here](#).

2. Select the “At Home Learner” button”. It should look like this:

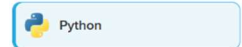
[At Home Learner](#)

3. Once directed to the new tab, select the “free” version and follow the instructions below to create an account.
4. You will be directed to your “My Section” page. Look at the top left of the page and click “Sandbox” to proceed.

Sandbox

5. Once you arrive at Sandbox, scroll down to the languages, select , then

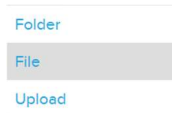
Python 3 Data Science (Graphics)
Use the powerful data science libraries numpy, pandas, an...



, and finally “Create Program”.

6. You have now launched the code workspace!

7. Upload your data file (preferably .txt, .csv may work as well) by clicking  located at the top left of the screen, the  button, and finally any one of the following:



Code to create a heatmap (without comments):

1. Copy and paste the following code into the python workspace:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

file_path = "package1.txt"

def plot_location_scan_grid(file_path):
    print(f"Reading data from {file_path}...")
```

```

df = pd.read_csv(file_path, sep=',', header=None, skiprows=1)
intensity_data = df.iloc[9371:9420, 1:50]
total_intensities = intensity_data.sum(axis=0).values
grid_data = total_intensities.reshape((7, 7))

fig, ax = plt.subplots(figsize=(10, 8))

cax = ax.imshow(grid_data, cmap='viridis')

cbar = fig.colorbar(cax)
cbar.set_label('Total Signal Intensity')

for i in range(7):
    for j in range(7):
        text_color = "white" if grid_data[i, j] < grid_data.max() / 2 else "black"
        ax.text(j, i, f"{grid_data[i, j]:.1e}",
                ha="center", va="center", color=text_color, fontsize=10)

ax.set_xticks(np.arange(7))
ax.set_xticklabels(['1', '2', '3', '4', '5', '6', '7'])

ax.set_yticks(np.arange(7))
ax.set_yticklabels(['1', '2', '3', '4', '5', '6', '7'])

plt.title('Intensity of 502-504 m/z by Location and Scan')
plt.xlabel('Scan Number')
plt.ylabel('Location Number')

plt.tight_layout()

plt.savefig('graph.png')
print("Success! Click on 'graph.png' in your left sidebar to view your graph.")

plt.show()

if __name__ == "__main__":
    plot_location_scan_grid(file_path)

```

Code to create heatmap with comments(Comments clarify the following code segment's functionality, represented by the # sign. They have no effect on your code.):

1. Copy and paste the following code into the python workspace:

```
#import statements (libraries of existing code)
```

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

#global variable that defines file path
file_path = "package1.txt"

#Procedure that extracts and plots data
def plot_location_scan_grid(file_path):
    print(f"Reading data from {file_path}...")

    #Panda reads data from the file and stores it in the variable "df"
    df = pd.read_csv(file_path, sep=',', header=None, skiprows=1)

    #Data from user-identified rows is extracted from "df" and stored in the
    #variable "intensity_data".
    #Numbers in the brackets are ranges for rows and columns, respectively.
    #Numbers in the brackets may be changed to your preference.
    intensity_data = df.iloc[9371:9420, 1:50]

    #Data within the row is added and displayed in the 7 x 7 graph
    total_intensities = intensity_data.sum(axis=0).values
    grid_data = total_intensities.reshape((7, 7))

    #Creates blank image and defines its dimensions (10in x 8 in)
    #Numbers may be changed to your preference
    fig, ax = plt.subplots(figsize=(10, 8))

    #Displays data as a heatmap
    cax = ax.imshow(grid_data, cmap='viridis')

    #Creates and labels a color bar using the min and max from your data
    cbar = fig.colorbar(cax)
    cbar.set_label('Total Signal Intensity')

    #The program loops through the 7 rows and 7 columns
    for i in range(7):
        for j in range(7):
            #Depending on the color of the square, the text is white or black, ensures
            #legibility of code.
            text_color = "white" if grid_data[i, j] < grid_data.max() / 2 else "black"
            #Data is extracted and rewritten in scientific notation
            ax.text(j, i, f"{grid_data[i, j]:.1e}",
                    ha="center", va="center", color=text_color, fontsize=10)

```

```

#Places data in specified positions on the x and y axis
ax.set_xticks(np.arange(7))
ax.set_xticklabels(['1', '2', '3', '4', '5', '6', '7'])

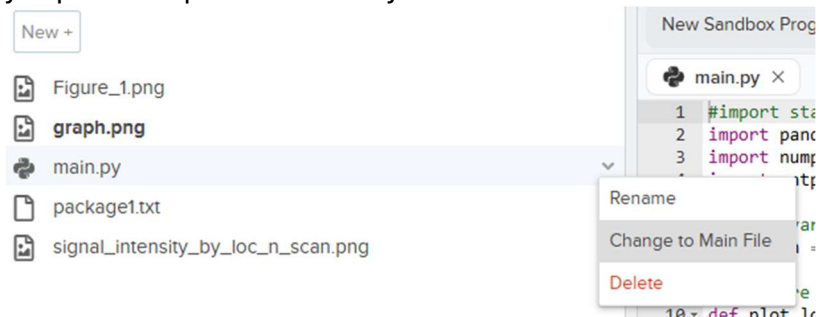
ax.set_yticks(np.arange(7))
ax.set_yticklabels(['1', '2', '3', '4', '5', '6', '7'])
#Labeling the heatmap, the x axis, and the y axis.
plt.title('Intensity of 502-504 m/z by Location and Scan')
plt.xlabel('Scan Number')
plt.ylabel('Location Number')
#Controls the margins
plt.tight_layout()

plt.savefig('graph.png')
print("Success! Click on 'graph.png' in your left sidebar to view your graph.")
#Displays the graph
plt.show()
#Somewhat like the procedure call. The spark to run the code
if __name__ == "__main__":
    plot_location_scan_grid(file_path)

```

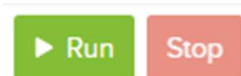
Troubleshooting:

- 1) Make sure that the tab that has your code (it may be named “main.py”) is your “main file”.
- 2) Click on the drop down bar of your main python code and select “Change to main file”. You may experience problems with your code if this is not enabled.



Running the code:

- 1) Click the green “Run” button located at the top left of your output screen.



Here is a link to my code: <https://codehs.com/sandbox/tealsheep3163/new-sandbox-program-4>